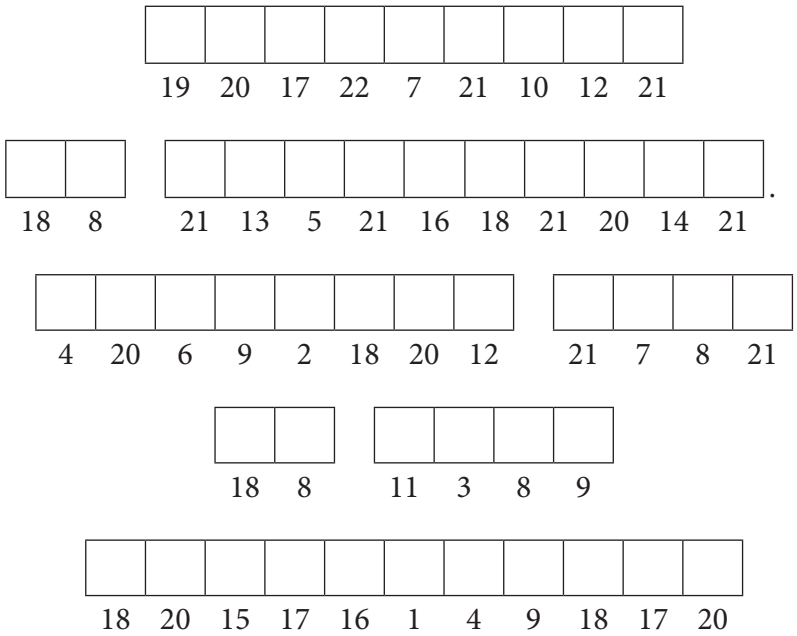
Name: _____

WORKSHEET

Equations of lines



Albert Einstein

Rules

The equation of a linear function can usually be determined given two pieces of information.

Gradient–*y*-intercept

$y = mx + c$

where m is the gradient and c is the y-intercept.

Gradient–point

$y - y_1 = m(x - x_1)$

where m is the gradient and (x_1, y_1) is the point given.

Two intercepts

$$\frac{x}{a} + \frac{y}{b} = 1$$

where $(a, 0)$ and $(0, b)$ are the x- and y-intercepts respectively.

Two points

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1} \text{ or } \frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

where (x_1, y_1) and (x_2, y_2) are the two points given.

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Questions

Find the equations of the following lines using the information given.

1 The gradient is 2 and the y-intercept is 3.

2 The line passes through $(-1, 7)$ and $(2, -2)$.

3 The gradient is 5 and the line passes through $(1, 3)$.

4 The x-intercept is $(7, 0)$ and the y-intercept is $(0, 7)$.

5 The line passes through $(-2, -9)$ and $(3, 11)$.

6 The gradient is -2 and the line passes through $(-5, 9)$.

7 The gradient is 1 and the y-intercept is $(0, 5)$.

8 The line has intercepts $(-6, 0)$ and $(0, 3)$.

9 The line passes through $(-1, 10)$ and $(2, 1)$.

10 The gradient is $\frac{2}{3}$ and the y-intercept $(0, -3)$.

11 The line passes through $(-2, 17)$ and $(3, -8)$.

12 The x-intercept is $\left(2\frac{1}{2}, 0\right)$ the y-intercept $(0, -5)$.

13 The gradient is 3 and the line passes through $(3, 14)$.

14 The line has intercepts $\left(1\frac{1}{4}, 0\right)$ and $(0, -5)$.

15 The line passes through $(1, -1)$ and $(3, -15)$.

16 The gradient is 2 and the y-intercept $(0, 0)$.

17 The x-intercept is $\left(-2\frac{2}{3}, 0\right)$ the y-intercept $(0, 8)$.

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18 The gradient is 4 and the line passes through (3, 12).

19 The line passes through $\left(2,\ 2\frac{1}{2}\right)$ and (8, 1).

20 The y-intercept is (0, 1) and the gradient is -5.

21 The gradient is 5 and the line passes through (0, 2).

22 The line has intercepts $(-3, 0)$ and (0, 3).

Solutions

A $y = -x + 7$

C $y = 4x - 5$

D $y = \frac{2}{3}x - 3$

E $y = 5x + 2$

F $y = -7x + 6$

G $y = 2x - 5$

H $y = -3x + 4$

I $y = 4x$

J $y = -5x + 7$

K $y = -\frac{1}{4}x + 3$

L $y = x + 5$

M $y = 2x + 3$

N $y = -5x + 1$

O $y = 3x + 8$

P $y = 4x - 1$

R $y = 2x$

S $y = \frac{1}{2}x + 3$

T $y = -3x + 7$

U $y = 5x - 2$

W $y = x + 3$

X $y = 3x + 5$

Y $y = -2x - 1$

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